MA 1250

THE UNIVERSITY OF WARWICK

FIRST YEAR EXAMINATION: JUNE 2013

INTRODUCTION TO GEOMETRY

Time Allowed: **1 hour**

Read carefully the instructions on the answer book and make sure that the particulars required are entered on each answer book.

Calculators are not needed and are not permitted in this examination.

ANSWER 2 QUESTIONS.

If you have answered more than the required 2 questions in this examination, you will only be given credit for your 2 best answers.

The numbers in the margin indicate approximately how many marks are available for each part of a question.

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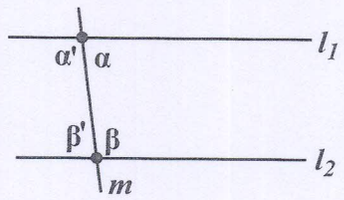
1. a) State Euclid's 5th Axiom. [2]

b) Let l_1 and l_2 be a pair of parallel lines. Let m be a line falling on l_1 and l_2. Let the internal angles m makes with l_1 and l_2 be as marked on the diagram



Prove that $\alpha' = \beta$ and $\alpha = \beta'$. [4]

c) Prove that the internal angles of a planar triangle sum to π. [3]

d) Let A and B be two points in the plane. By making use of the standard criteria for congruence of triangles or otherwise, prove that the perpendicular bisector of AB is the locus of points equidistant from A and B. [6]

e) Let $\triangle ABC$ be a triangle in the plane. Let l_{AB} and l_{BC} be the perpendicular bisectors of AB and BC, respectively. Show that l_{AB} and l_{BC} cannot be parallel, and therefore intersect. Show that their point of intersection lies on the perpendicular bisector of AC. [10]

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2. Let S be a sphere in the ambient Euclidean 3-space. Let O denote its center.

a) Let A and B be two points on S. Define two notions of distance between A and B - the intrinsic one and the ambient one. [4]

b) Let $f\colon S \to S$ be a map from S to itself. What does it mean to say that f is an isometry? Does it matter which of the above two notions of distance to use when answering this question? [2]

c) When does an isometry of the ambient 3-space restrict to an isometry of the sphere S? Justify your answer. [3]

d) For each of the following isometries of the ambient 3-space, explain when (and why) does it restrict to an isometry of the sphere S:

(i) Rotation around a line. [2]

(ii) Reflection in a plane. [2]

(iii) Translation by a vector. [2]

e) Prove that if AB and CD are arcs of great circles on S which are of the same length, then there exists an isometry of the sphere which takes A to C, B to D and the arc AB to the arc CD. You can use the fact that any two distinct planes passing through the center O of S intersect along a single straight line. *Hint:* First find a rotation sending one great circle to the other. [10]

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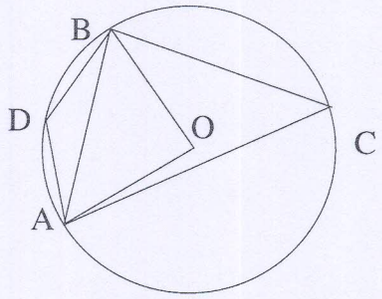
3. a) State the three criteria for triangles $\triangle ABC$ and $\triangle A'B'C'$ to be congruent. [6]

b) Let $\triangle ABC$ be a triangle in the plane.

(i) What is meant by saying that $\triangle ABC$ is an isosceles triangle? [1]

(ii) Prove that $\triangle ABC$ is an isosceles triangle if and only if two of its interior angles are equal. [4]

c) Let S be a circle with center O and let A, B, C and D be as on the figure below:



(i) Prove that $\angle BOA = 2\angle BCA$. [6]

(ii) Prove that $2\pi - \angle BOA = 2\angle ADB$. [8]

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